

Elena Cossu, PhD

EUI MAX WEBER FELLOW AND POSTDOCTORAL RESEARCHER

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Data scientist and political scientist with a background in computational social science, specializing in Natural Language Processing (NLP) and data science applications to comparative European political economy. Currently a Max Weber Fellow at the European University Institute, developing NLP-based tools to predict economic preferences in Europe at the Robert Schuman Centre. Previous experience includes a postdoctoral position at Sciences Po Paris, where I applied NLP methods to map and forecast party configurations across Europe, and work as a Senior Policy Evaluation Researcher at the UK National Centre for Research, assessing the impact of education policies. Extensive expertise in quantitative social research, with a focus on the intersection of data science and comparative political research.

Employment

European Central Bank (ECB)

DATA SCIENTIST

Frankfurt, Germany

June 2026 - Present

European University Institute

MAX WEBER FELLOW

Florence, Italy

Sept. 2025 - June 2026

- Affiliated to the Robert Schuman Centre for Advanced Studies
- Using NLP to build a tool to predict political preferences in Europe

SciencesPo Paris

POSTDOCTORAL RESEARCHER FOR THE AUTHLIB PROJECT

Paris, France

Apr. 2023 - Aug. 2025

- Affiliated to the Centre for European Studies and Comparative Politics
- Data collection and modelling for the project “Neo-authoritarianisms in Europe and the liberal democratic response” (AUTHLIB), an Horizon Europe and funded by the European Commission
- Using Natural Language Processing (NLP) to map political ideological configurations and dimensions across Europe
- Working under the direction of Prof. Rovny and Prof. Froio at the CEE and Prof. Cointet at Medialab

National Centre for Social Research

SENIOR POLICY EVALUATION RESEARCHER

London, United Kingdom

Nov. 2021 - Oct. 2024

- Design and implementation of surveys and survey experiments to measure the effect of policies linked to education
- Design and implementation of experiments (Randomised Control Trials) and Quasi-Experiments (e.g. Synthetic Control, Propensity Score Matching, etc.) to assess causality
- Clients include the UK Ministry of Justice, The National Lottery Community Fund, the UK Department for Education, the Education Endowment Foundation, the Department for Health and Social Care (DHSC), and others

Corvinus University of Budapest

ASSISTANT LECTURER

Budapest, Hungary

Sep. 2021 - Jan. 2022

- Assistant teacher for Master level courses

European Commission MSCA

EARLY STAGE RESEARCHER

Budapest and London

Aug. 2018 - Sep. 2021

- Researcher as part of the Horizon 2020 European Commission project FATIGUE, a Marie Skłodowska-Curie Innovative Training Network (MSCA)
- Assist with the quantitative analysis and the writing of reports and articles for long-term research projects
- Duty stations for the project: (1) the European Bank for Reconstruction and Development (EBRD), Office of the Chief Economist, Transition Impact and Global Economics Team; (2) the Slavonic School of Eastern European Studies (UCL SSEES); (3) Corvinus University of Budapest Institute of World Economy

Education

Corvinus University and University College London

PHD IN ECONOMICS

Budapest and London

Aug. 2018 - Oct. 2023

- Dissertation title: ‘Essays on Conceptualising Policies Linked to Populist Rhetoric in Europe and Measuring their Effects in Hungary’
- Co-supervised at University College London (UCL)
- PhD sponsored by the European Commission Marie Skłodowska-Curie Innovative Training Network (MSCA) as part of the FATIGUE research project

- MA in International Public Management with a specialisation in Global Risks from SciencesPo Paris School of International Affairs (PSIA), with a final dissertation called «Fighting terrorism and fundamental rights»
- MSc in Economics and Management of International and Public Affairs from Bocconi University, with a dissertation called «The Parisian transport system: a best practice of urban mobility»

Università Cattolica del Sacro Cuore

- Final dissertation: «The Hungarian Transition from Planned to Capitalist Economy»
- Erasmus at Pázmány Péter Catholic University in Budapest during the 2013/2014 academic year

Peer-Reviewed Publications

How Critical Events Shape Political Agendas on Multiculturalism in France

Politics and Governance

<https://www.cogitatiopress.com/politicsandgovernance/article/view/9743>

- 2025 This study investigates how terror attacks influenced political discourse on multiculturalism in France between 2014 and 2021. By analysing 143,870 tweets from major political parties and using a Natural Language Inference model to label them, the research explores how far-right discourses influenced mainstream political communication, particularly concerning issues related to multiculturalism the accommodation of immigration and cultural diversity. We track the salience and sentiment of multiculturalism-related discussions across political parties by using a XLM-RoBERTa model using Natural Language Inference to classify texts.

Application of Natural Language Processing to the Electoral Manifestos of Parties Characterised by Populist Rhetoric in Central and Eastern Europe

East European Journal of Society and Politics

<https://bit.ly/42TL65F>

- 2025 This study applies a variety of NLP techniques to analyze electoral manifestos of populist parties in Central and Eastern Europe, uncovering key rhetorical patterns and ideological shifts. It finds a consistent use of cleavages around words such as Europe and corruption, highlighting in a quantitative way how discourse is use to exploit certain cleavages in political narratives.

Clustering and Analysing Relevant Policy Dimensions of Populist, Left-Wing, Centrist, and Right-Wing Parties across Europe

Central European Journal of Public Policy, 17(1), 41–54

<https://doi.org/10.2478/cejpp-2023-0004>

- 2023 This paper is a configurational analysis that creates a new theoretical elaboration of populist parties in Europe in terms of policy choices. To do so, the paper analyses policy positions connected to the populist literature for 242 parties in Europe in 2014 and 2019 using the Chapell Hill Expert Survey and the PopuList Survey data. Groups of parties with similarly held positions in 2014 and 2019 are identified using cluster analysis. The result highlights that rather than across a classical right-left divide connotation, Europe can be divided into four clusters: right-wing populists, a moderate pro-Europe left, a pro-Europe pro-liberalism centre and an intersectional left based on identity politics.

The Economic Effect Of Populist Rhetoric In Hungary

Online Journal of Modelling the New Europe, 41, 105–138

<https://doi.org/10.24193/OJMNE.2023.41.05>

- 2023 This paper studies the effect of populist rhetoric on GDP in Hungary from 2010 to 2023 using three methodologies: Difference-in-Differences, Synthetic Difference-in-Differences, and the Synthetic Control Method.

Understanding Contemporary Populism Through the Latin American Experience

AARMS – Academic and Applied Research in Military and Public Management Science, 20(3), Article 3

<https://doi.org/10.32565/aarms.2021.3.4>

- 2021 This paper discusses how the Latin American experience can help us understand contemporary populism in Europe. To do so, we look at the state of the literature on populism, its relation to the Latin American experience, the evolution of the approach to populism, and the conclusions we can draw from these different perspectives. We conclude that contemporary populism is also limited in the same way the contextual approach to Latin American populism was limited. This also helps us understand why we still do not have a shared definition of populism.

The Gravity Model for Trade Theory

Köz-gazdaság, 14(3), Article 3

<https://doi.org/10.14267/RETP2019.03.25>

- 2019 This article delivers a theoretical overview of the gravity model and some of its main applications. The goal is as well to provide some criticism and to discuss the effectiveness of this widely used model. The model, in fact, not only did not lose its appeal over time but, on the contrary, it continued to develop and to become more reliable and relevant. It becomes clear that, although it has some problems and limitations, the gravity model is still a widely used and compelling tool, especially in the trade flow area.

Book Chapters

The effect of policies of Populist Parties in power on economic performance: The case of Hungary

Cossu, E.

- 2025 In J. Kubik & R. C. M. Mole (Eds.), *The Anatomy of Right-Wing Populism: Dealing with Transformational Fatigue in Central and Eastern Europe* (Chapter 6). Available at:
<https://uclpress.co.uk/book/the-anatomy-of-right-wing-populism/>

Under Peer-Review

Diversity and Democracy: an EconML approach

Submitted to *Comparative Political Studies*

Rovny, J., Cossu, E., & Sattelmayer, L.

- 2025 Can diversity and democracy coexist? This article studies the democratic support of Muslim minorities in western Europe. Departing from past studies, it argues that ostracized ethnic minorities – viewed as a cultural or political threat by significant portions of the majority – have an incentive to support democratic institutions in order to seek legal equality and civil liberties. This search for democracy is attenuated by religious particularism and cultural conservatism. Using European Social Survey (ESS Round 10) data, we pair a transparent linear regression baseline with EconML's Double Machine Learning estimators (LinearDML and DR Learner) to target causal effects under ignorability and overlap, and we visualize within-group heterogeneity with conditional inference trees. The article demonstrates that more secular or liberal Muslim minorities are significantly more likely to support democracy than the majority. Religious or conservative Muslims do not differ from the majority. These findings suggest that, overall, western European Muslims are an asset for liberal democracy.

Works in Progress

Blurring the Boundaries: Conceptualizing and Measuring Illiberal and Far-Right Party Communication in Europe

Cossu, E., Froio, C. & Völker, T.

- 2025 This paper maps illiberal elements in the communication of mainstream European far right parties using Twitter data. We collect all tweets posted between 2006 and 2023 by the official accounts of these parties and, when relevant, by their leaders while in office within the party. Using BERT, each tweet is classified along four dimensions: nativism, authoritarianism, populism, and references to liberal democratic institutions such as checks and balances, judicial independence and media freedom. We then examine how often the core far right repertoire of nativism, authoritarianism and populism co occurs with institutional discourse. This enables us to estimate the degree to which far right communication also articulates illiberal discourse and to identify variation over time and across parties.

A Large Language Model Framework for Creating and Evaluating Political Data

Cossu, E.

- 2025 This paper introduces a novel framework to label and evaluate large datasets for complex political science concepts. It does so by substituting traditional labelling and supervised machine learning models with an open-source models that uses and evaluates different open source large language models against a small sample of examples for any given concept. The labelling is also evaluated against a confidence interval and Rouge-L, BLEU and BertScore metrics. Thanks to BertViz visualization the model also helps researchers to assess why the LLMs labeled concepts a certain way, giving human interpretability to the findings.

Reports

How Terror Attacks Pushed French Politics Closer to the Far Right

Authlib Project and SciencesPo Paris

2026 <https://www.authlib.eu/how-terror-attacks-pushed-french-politics-closer-to-the-far-right/>
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Literature Review on How to Measure Illiberalism Using Text Data

Authlib Project and SciencesPo Paris

2023 <https://www.authlib.eu/literature-review-how-to-measure-illiberalism-using-text-data/>
This literature review explores different approaches to measuring illiberalism in political texts using Natural Language Processing (NLP) techniques. It provides a framework for analyzing democratic backsliding and illiberal discourse in contemporary politics, especially for political scientists.

Yorkshire and Humber COVID-19 campaign pilot

UK Health Security Agency

2023 <https://assets.publishing.service.gov.uk/media/652fe1fc92895c000ddcb9c3/Yorkshire-and-Humber-COVID-19-campaign-pilot-evaluation-report.pdf>
This pilot study evaluates the impact of COVID-19 public health messaging in Yorkshire and Humber, focusing on behavioral responses and policy effectiveness in increasing vaccine uptake and public safety measures. It uses Synthetic Control Methods as a methodology.

LNK Educate: Feasibility and pilot study

Youth Endowment Foundation

2023 <https://youthendowmentfund.org.uk/funding/evaluations/lmk-educate/>
An evaluation of the LNK Educate program, assessing its feasibility and effectiveness in reducing youth violence and improving educational outcomes through targeted interventions using a before-after design.

Other Publications

Populists in Europe: A Political Group with Its Distinct Policy Preferences?

AuthLIB Blog

2024 <https://www.authlib.eu/populists-in-europe-a-political-group-with-its-distinct-policy-preferences/>

A Small Country Trying To Go Big, A Big Union Trying Not To Go Small

Australian Institute of International Affairs

2022 <https://www.internationalaffairs.org.au/australianoutlook/a-small-country-trying-to-go-big-a-big-union-trying-not-to-go-small/>

Chinese vaccine diplomacy in Latin America could change the international order

ECPR The Loop

2021 <https://theloop.ecpr.eu/chinese-vaccine-diplomacy-in-latin-america-could-change-the-international-order/>

Book Review: The Glass Half-Empty: Debunking the Myth of Progress in the Twenty-First Century by Rodrigo Aguilera

LSE Review of Books

2020 <https://blogs.lse.ac.uk/lsereviewofbooks/2020/10/27/book-review-the-glass-half-empty-debunking-the-myth-of-progress-in-the-twenty-first-century-by-rodrigo-aguilera/>

Grants and Funding

2023 AUTHLIB – Neo-Authoritarianisms in Europe and the Liberal Democratic Response, European Union and the UK Research and Innovation

2018 Delayed Transformational Fatigue in Central and Eastern Europe, Horizon 2020 Grant agreement ID: 765224

Awards and Honors

ACRI Young Investigator Prize

ECOMOD 2021 (2023)

2023 Awarded for outstanding contributions in economic modeling, recognizing innovative research in applied computational economics.

Presentations and Conferences

- 2026 **ECPR Joint Sessions 'From Code to Conversation: What LLMs Mean for Political Science' Workshop A** Framework for Labelling Complex Political Concepts Using Open-Weight LLMs: Evidence from Illiberalism in Party Communication
- 2025 **International Political Science Association (IPSA) Conference** How Critical Events Shape Political Agendas on Multiculturalism in France
- 2025 **European Political Science Association (EPSA) Conference** The Social Bases of Democracy in Europe
- 2025 **Third Annual CIVICA Political Behaviour and Institutions Conference (PoBI)** Measuring Illiberalism Using Text Data and Machine Learning
- 2025 **ECPR Joint Sessions: Text-as-Data Analysis in Political Science: Unlocking Political Insights with New Research Tools** Measuring Illiberalism Using Text Data and Machine Learning
- 2025 **Helsinki Conference on Emotions, Populism, and Polarisation (HEPP5)** Measuring Illiberalism Using Text Data and Machine Learning
- 2024 **MethodsNet Conference** Measuring Illiberalism Using Text Data and Machine Learning
- 2022 **ECPR General Conference** Assessing the economic performance of a government characterised by populist rhetoric in Hungary: A synthetic control approach
- 2021 **The Doctoral Student Association of the University of Pécs Conference (PTE DOK)** Macroeconomics and Politics in Hungary: How an Illiberal Government Characterised by Populist Rhetoric Deals With its Public Budget
- 2021 **inGRID Workshop** Macroeconomics, Policies, and CEE: a Machine Learning Approach to Social Investment and Social Protection Policies
- 2020 **ASEEES 52nd Annual Convention** A Review of Quasi-Experimental and Predictive Methodologies in Relation to Measuring Populism and its Outcomes
- 2020 **Science is Wonderful! European Commission Outreach Event** FATIGUE Project Workshop
- 2019 **WINIR Symposium 2019** Economic populism, fiscal policies and budget deficits: the sustainability implications for Central and Eastern Europe
- 2019 **RSA Winter Conference** Present and future populist policies in Europe: the Cases of Hungary and Bulgaria
- 2019 **UACES 49th Annual Conference** The New Populist Policies Prescriptions

Teaching experience

- 2026 S1 **Populism and Illiberalism in Europe: Tracking Opinions and Measuring Consequences** (Freie Universität Berlin (FU))
Guest teacher for the course 'Populism: How to Study a Contested Concept' of Professor Bruno Castanho Silva and as part of the Max Weber Fellow Teaching Week.
- 2026 S1 **LLM Techniques, Text Analysis and Illiberal Speech Methods Seminar** (Scuola Normale Superiore (SNS))
In this methodological workshop for PhD researchers, I taught advanced approaches to the analysis of political text using natural language processing and large language models. The session provided a hands-on introduction to the full research pipeline, from data collection via web scraping to the application of pre-trained models, BERT-based classification, and generative LLMs. Participants engaged with real-world applications, including the detection of illiberal and anti-democratic discourse in political communication, and were equipped with practical skills in Python and the Hugging Face ecosystem. The course emphasized both methodological rigor and critical reflection on the ethical and substantive challenges of measuring democratic backsliding through text-as-data approaches.
- 2024 S1 **Policy Analysis and Assessment Methods Course** (SciencesPo Paris)
The Policy Analysis and Policy Assessment course is a course for master's students from various schools across SciencesPo. In this course, I taught students the theory, practice, and examples of the most widely used methodologies in industry for evaluating the effects of policies. The course covered a broad range of methods, from interviews to advanced machine learning techniques. Students were also provided with examples in R and Python to equip them with the skills needed to apply these methodologies in their future careers.

Democracy in Europe Seminar (SciencesPo Paris)

2024 S1 In this seminar I thought the political economy of populism in Hungary, looking extensively at the drivers and effects of populist rhetoric in the country.

Democracy in Europe Seminar (SciencesPo Paris)

2023 S2 An in-depth examination of democratic institutions and their resilience in European politics, with case studies from different EU member states.

Text as Quantitative Data PhD Seminar (SciencesPo Paris)

2023 S2 This seminar introduces computational methods for text analysis, including NLP techniques for political and social science research.

Impact Evaluation Quasi-Experimental Design (NatCen Course for the Office for National Statistics (ONS))

2023 S2 An applied course on quasi-experimental methods for evaluating policy interventions, with practical examples from public policy research for the workers of the Office of National Statistics in the United Kingdom.

Impact Evaluation Quasi-Experimental Design (NatCen Course for the Office for National Statistics (ONS))

2022 S2 A methodological training on how to assess the causal impact of policies using econometric and statistical approaches for the workers of the Office of National Statistics in the United Kingdom

International Organizations and Economic Diplomacy Seminar (Corvinus University of Budapest)

2021 S1 The course examined the role of international organizations in global economic governance and their influence on economic diplomacy.

Global Economic Trends Seminar (Corvinus University of Budapest)

2021 S1 An overview of key trends in the global economy, including financial crises, trade dynamics, and economic policy shifts.

Development Economics Seminar (Corvinus University of Budapest)

2018 S2 This course covers the main theoretical and empirical debates in development economics, including poverty alleviation and economic growth models.

Professional Service

2023 Blind peer-reviewer for East European Politics and Societies

Relevant Skills

Data Analysis

- Natural Language Processing (NLP)
- BERT Modelling
- LLMs fine-tuning
- Web scraping
- Quantitative Multi-Lingual Sentiment Analysis
- Cluster Analysis
- Survey Design
- Imputation Techniques (K+means and regression analysis inputting)
- Randomized Control Trials (RCTs)
- Matching Methods (including Mahalanobis matching and Kernel matching)
- Propensity Score Weighting (PSW)
- Synthetic Control Methods (SCM)

Programming Languages:

- R, Python, SQL, command-line programming, Git, STATA, SPSS, HTML, LaTeX

Languages:

- English, French, Italian, Spanish, Hungarian (basic)

Certifications

2023 **Google** Data Analytics Professional Certificate
<https://www.coursera.org/account/accomplishments/verify/XZCBSTMZ3W4>
2022 **Office for National Statistics (UK)** Accredited Researcher
2019 **Centre for Microdata Methods and Practice** Policy Evaluation Methods

References

- Professor Jan Rovny, Sciences Po Paris (CEE), jan.rovny@sciencespo.fr
- Professor Caterina Froio, Sciences Po Paris (CEE), caterina.froio@sciencespo.fr